

Use-Case Flow Specification

Problem Statement #48 — Incident Escalation & On-Call Rotation Engine | Lab 1: Requirements Engineering & UML Use-Case Modelling

Use Case ID:	UC-04 (with UC-01)	Primary Actor(s):	On-Call Engineer, Secondary On-Call Engineer
Use Case Name:	Handle and Escalate Incident Alert	Secondary Actor(s):	Monitoring System, Notification Gateway, Incident Commander

Preconditions

- The Monitoring System is integrated with the platform via a registered webhook endpoint.
- An on-call rotation schedule is configured and contains a currently active primary and secondary engineer for the affected service.
- The On-Call Engineer and Secondary On-Call Engineer have verified, reachable contact channels (push, SMS, email) on file.

Postconditions

- Success: The incident is marked "Acknowledged," the acknowledging engineer and timestamp are recorded, and the Incident Commander dashboard reflects the current escalation tier.
- Failure: The incident remains "Unacknowledged," having been escalated through all tiers, and is flagged for manual Incident Commander intervention.

Main Success Scenario

- 1 The Monitoring System sends an incident alert (service, severity, timestamp) to the system via webhook.
- 2 The system validates the alert payload and creates a new Incident record with a unique incident ID.
- 3 The system queries the on-call rotation schedule (UC-02, *included*) to identify the active primary On-Call Engineer for the affected service.
- 4 The system dispatches a notification (UC-03, *included*) to the primary On-Call Engineer via push, SMS, and email within 3 seconds.
- 5 The On-Call Engineer receives the notification and acknowledges the incident within the 5-minute acknowledgment window.
- 6 The system timestamps the acknowledgment, updates the incident status to "Acknowledged," and records the acknowledging engineer.
- 7 The system updates the Incident Commander dashboard with the current incident status. The use case ends successfully.

Alternate Flow — 5a: Unacknowledged Timeout (Escalation, UC-04 extends UC-01)

- 5a1. Five minutes elapse from dispatch (step 4) without acknowledgment from the primary On-Call Engineer.
- 5a2. The system automatically escalates the incident, dispatching a notification to the Secondary On-Call Engineer via SMS and a voice call.
- 5a3. If the Secondary On-Call Engineer also fails to acknowledge within a further 5 minutes, the system escalates directly to the Incident Commander and marks the incident "High-Risk — Unacknowledged."
- 5a4. Once any escalation tier acknowledges the incident, the use case resumes at step 6 of the main success scenario.